

Lecture 5: Aug 27

Last time

- Axiomatic Foundations (1.2)

Today

- Calculus of Probabilities (1.2)

Definition Given a sample space S and an associated sigma algebra $\mathcal{B}$, a *probability function* is a function $\Pr$ with domain $\mathcal{B}$ that satisfies

1. $\Pr(A) \geq 0$ for all $A \in \mathcal{B}$.
2. $\Pr(S) = 1$.
3. If $A_1, A_2, \dots \in \mathcal{B}$ are pairwise disjoint, then $\Pr(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \Pr(A_i)$.

The above three properties are usually referred to as the Axioms of Probability (or the Kolmogorov Axioms, after A. Kolmogorov, one of the fathers of probability theory). Any function that satisfies the Axioms of Probability is called a probability function.

Example Consider the simple experiment of tossing a fair coin (just once), so $S = \{H, T\}$. A reasonable probability function is the one that assigns equal probabilities to heads and tails, that is,

$$\Pr(\{H\}) = \Pr(\{T\}).$$

Since $S = \{H\} \cup \{T\}$, we have, from Axiom 2, $\Pr(\{H\} \cup \{T\}) = 1$. Also, $\{H\}$ and $\{T\}$ are disjoint, so $\Pr(\{H\} \cup \{T\}) = \Pr(\{H\}) + \Pr(\{T\})$. Collectively, we have

$$\begin{aligned}\Pr(\{H\}) &= \Pr(\{T\}) \\ \Pr(\{H\} \cup \{T\}) &= 1 \\ \Pr(\{H\} \cup \{T\}) &= \Pr(\{H\}) + \Pr(\{T\})\end{aligned}$$

Therefore, $\Pr(\{H\}) = \Pr(\{T\}) = \frac{1}{2}$.

Calculus of Probabilities

We start with some fairly self-evident properties of the probability function when applied to a single event.

Theorem If $\Pr$ is a probability function and A is any set in $\mathcal{B}$, then

1. $\Pr(\emptyset) = 0$, where $\emptyset$ is the empty set;
2. $\Pr(A) \leq 1$;
3. $\Pr(A^c) = 1 - \Pr(A)$.

proof:

Theorem If $\Pr$ is a probability function and A and B are any sets in $\mathcal{B}$, then

1. $\Pr(B \cap A^c) = \Pr(B) - \Pr(A \cap B)$;
2. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$;
3. If $A \subset B$, then $\Pr(A) \leq \Pr(B)$.

proof:

Formula (2) in the above theorem gives a useful inequality for the probability of an intersection (Bonferroni's Inequality):

$$\Pr(A \cap B) \geq \Pr(A) + \Pr(B) - 1.$$

Theorem If $\Pr$ is a probability function, then

1. $\Pr(A) = \sum_{i=1}^{\infty} \Pr(A \cap C_i)$ for any partition $C_1, C_2, \dots$;
2. $\Pr(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \Pr(A_i)$ for any sets $A_1, A_2, \dots$.

where (1) is also referred to as "Total probability" and (2) is Boole's inequality.

proof:

Theorem If $\Pr$ is a probability function, then

1. $\Pr(A) = \sum_{i=1}^{\infty} \Pr(A \cap C_i)$ for any partition $C_1, C_2, \dots$;
2. $\Pr(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \Pr(A_i)$ for any sets $A_1, A_2, \dots$.

where (1) is also referred to as "Total probability" and (2) is Boole's inequality.

proof:

Conditional Probability

All of the probabilities that we have dealt with thus far have been unconditional probabilities. A sample space was defined and all probabilities were calculated with respect to that sample space. In many instances, however, we are in a position to update the sample space based on new information. In such cases we want to be able to update probability calculations or to calculate *conditional probabilities*.

Definition If A and B are events in S , and $\Pr(B) > 0$, then the *conditional probability* of A given B , written $\Pr(A|B)$, is

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

Note that B becomes the sample space now: $\Pr(B|B) = 1$.